Linear Algebra

Vector Spaces



Series

5 December 2022

Contents

1 Euclidean Vector Spaces **1**

1.1 Vector Spaces . . . 1

1.1.1 Examples . . . 3

1.1.2 Example 1 . . . 3

1.2 Linear Combinations . . . 3

1.2.1 Norm, Dot Product, and Distance In $\mathbb{R}^n$ . . . 4

1.2.2 Examples . . . 5

Example 1 . . . 5

1.2.3 Unit Vectors . . . 5

Example: Normalizing A Vector . . . 5

1.2.4 The Standard Unit Vectors . . . 5

Example: Linear Combination of Standard Unit Vectors . . . 6

1.2.5 Distance In $\mathbb{R}^n$ . . . 6

Example: Calculating distance in R^n . . . 7

1.3 Inner Products . . . 7

1.3.1 Dot Product . . . 7

Example: Find The Dot Product . . . 8

1.3.2 Component Form of Dot Product . . . 9

Example: Geometry Problem Solved Using Dot Product . . . 10

1.3.3 Algebraic Properties of The Dot Product . . . 11

Calculating With Dot Products . . . 11

Orthogonality . . . 12

1.3.4 Lines and Planes Determined By Points and Normals . . . 12

Example: Point Normal Equations . . . 13

Example: Point Normal Equations In 3D . . . 13

Example: Vectors Orthogonal To Lines and Planes Through Origin . . . 14

1.3.5 Decomposing A Vector Into A Sum of Two Orthogonal Vectors 14

Example: Orthogonal Projection on a Line . . . 16

Example: Vector Component of u Along a . . . 17

1.3.6 Defining Projections . . . 17

1.3.7 Distance In $\mathbb{R}^n$ . . . 17

Example: Distance Between A Point And A Plane . . . 19

Example: Distance Between A Point And A Plane (2) 19
Example: Distance Between Parallel Planes 20

2 Geometry of Linear Systems **21**
2.1 Linear Systems 21
2.1.1 Parameterization 22
Example: Vector and Parametric Equations of Lines in $\mathbb{R}^2$ and $\mathbb{R}^3$ 23
Example: Vector and Parametric Equations of a Plane in $\mathbb{R}^3$. 23
Example: Vector and Parametric Equations of Lines and Planes in $\mathbb{R}^4$ 24
2.1.2 Lines Through Two Points In $\mathbb{R}^n$ 25
Example:A Line Through Two Points in $\mathbb{R}^2$ 25
2.1.3 Line Segment From One Point To Another 26
Example: A Line Segment from One Point to Another in $\mathbb{R}^2$. 26
2.1.4 Parametric Equations of Lines 26
Example: Find parametric equations of the line 27
Example: Find parametric equations of the line in $\mathbb{R}^3$ 28
2.2 Planes In 3-Space 28
2.2.1 Determining Parallel Planes and Equations of Planes 29
Example: Determine Whether The Planes Are Parallel 29
Example: Finding Equations of Plane Through Points and Normal 29
Example: Finding Equations of Plane Through Points 30
Example: Determining Whther Line Is Parallel To Plane . . . 30
Example: Finding Intersection Between Line and Plane 30
Example: Finding Intersection Between Line and Plane (2) . . 31
Example: Finding The t Parameter Given Normal, Points and Vector 33
2.2.2 Intersecting Planes 33
Example: Finding Angles of Intersection 33
Example: Finding Line Equation of Angles of Intersection . . 34

3 Coordinatization **36**
3.1 Coordinates 36
3.1.1 Change of Coordinates By Rotation 36
3.1.2 Extending To $\mathbb{R}^3$ 37
3.1.3 Change of Basis Transformation 37
3.1.4 Change of Coordinates By Translation 38
3.1.5 Change of Basis By Translation 38
3.1.6 Cylindrical Coordinates 39
Example: Convert Cartesian Coordinates To Cylindrical Coordinates 40
3.1.7 Spherical Coordinates 40

Example: Convert From Cylindrical Coordnates To Spherical Coordinates 42
Example: Convert From Cartesian Coordinates to Spherical Coordinates 42

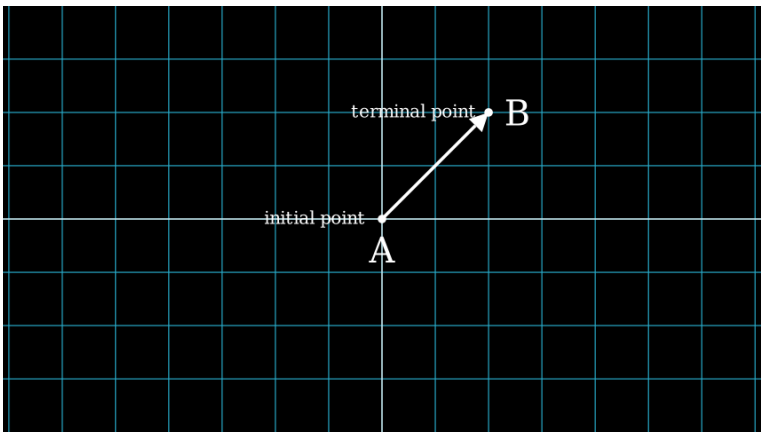
4 Cross Product **44**

Chapter 1

Euclidean Vector Spaces

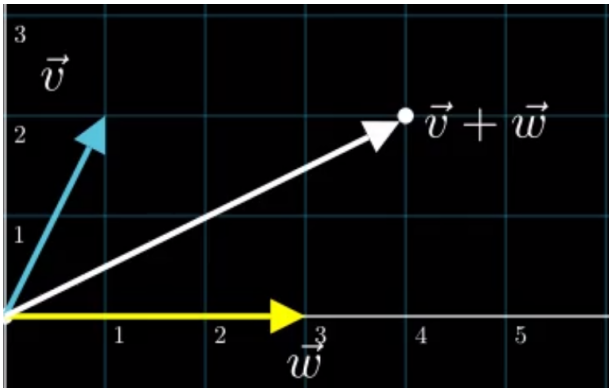
1.1 Vector Spaces

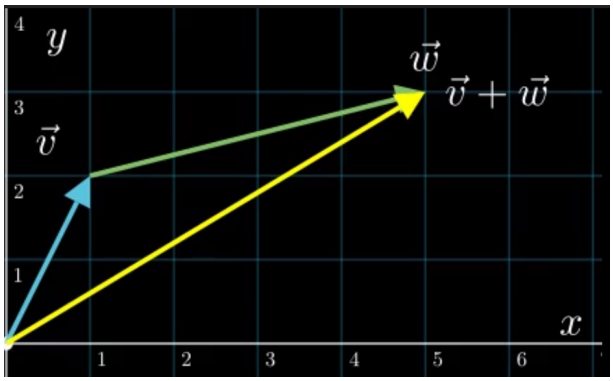
Scalar: Quantities that can be described by numerical value alone.
Vector: Quantities that require both number and direction for physical description.
Geometric Vectors



$$\vec{v} = \vec{AB}$$

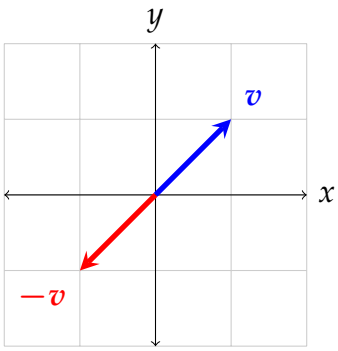
Triangle rule for vector addition:
If $\vec{v}$ and $\vec{w}$ are vectors in 2-space or 3-space that are positioned so the initial point of $\vec{w}$ is at the terminal point of $\vec{v}$





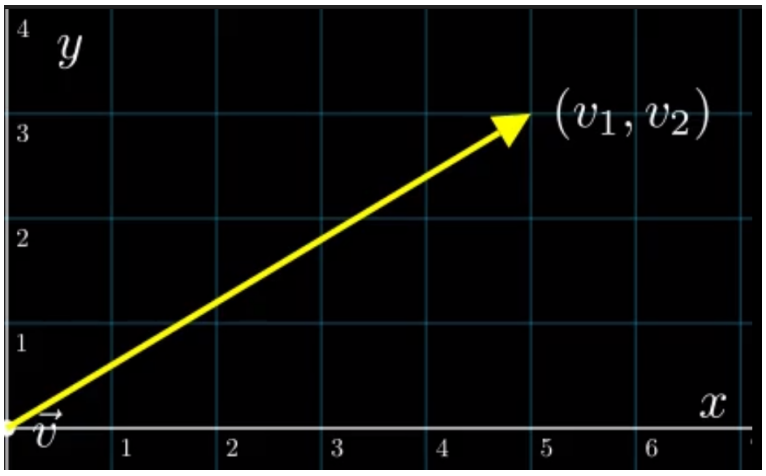
Vector Subtraction

$$\vec{w} - \vec{v} = \vec{w} + (-\vec{v}) \tag{1.1}$$

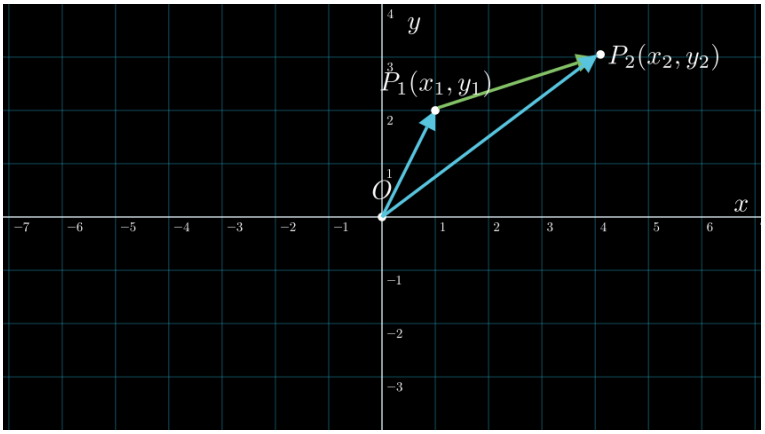


Collinear: Vectors that lie on the same line.
If a vector $\vec{v}$ in 2D or 3D space is positioned with its initial point at the origin of a rectangular coordinate system, then the vector is completely determined by the coordinates of its terminal point.

We call these coordinates the components of v relative to the coordinate system.



If $\vec{P_1P_2}$ denotes the vector with initial point $P_1(x_1, y_1)$ and terminal point $P_2(x_2, y_2)$ then the components are



$$\begin{aligned} \vec{P_1P_2} &= (x_2 - x_1, y_2 - y_1) \\ &= \vec{OP_2} - \vec{OP_1} \\ &= (x_2, y_2) - (x_1, y_1) = (x_2 - x_1, y_2 - y_1) \end{aligned}$$

1.1.1 Examples

1.1.2 Example 1

Find the components of a vector. The components of the vector

$\vec{v} = \vec{P_1P_2}$ with **initial point** $\vec{P_1}(2, -1, 4)$ and **terminal point** $\vec{P_2}(7, 5, -8)$ are

$$\vec{v} = (7 - 2, 5 - (-1), -8 - 4) = (5, 6, -12)$$

Consider $\mathbb{R}^2$, the superscript reinforces the idea that the ordered pairs correspond to points in the plane.

1.2 Linear Combinations

Definition 1.2.1 (Definition 1)**.** If n is a positive integer, then an ordered n-tuple is a sequence of n real numbers $(v_1, v_2, \ldots, v_n)$. The set of all ordered n-pairs is called n-space and it is denoted by $\mathbb{R}^n$

Definition 1.2.2 (Definition 2)**.** If $\vec{w}$ is a vector in $\mathbb{R}^n$, then $\vec{w}$ is a linear combination of the vectors $v_1, v_2, \ldots, v_n$ in $\mathbb{R}^n$ if it can be expressed in the form.

$$w = k_1v_1 + k_2v_2 + \cdots + k_nv_n$$

where k_1, k_2 are scalars. These scalars are called coefficients of the linear combinations

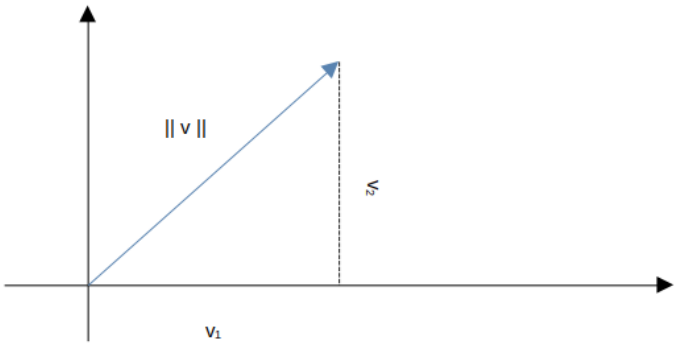
1.2.1 Norm, Dot Product, and Distance In $\mathbb{R}^n$

Definition 1.2.3 (Definition 3)**.** The **Norm**, otherwise known as the *length*, of a vector $\vec{v}$ is defined as

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2}$$

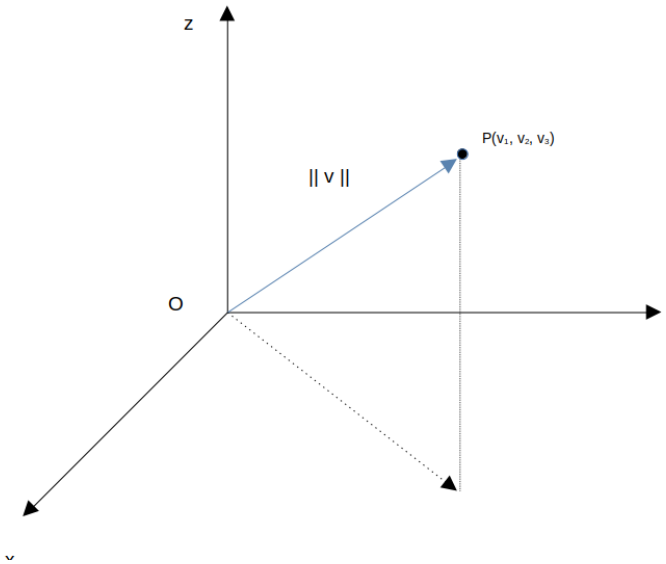
Norm In $\mathbb{R}^2$

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2} \tag{1.2}$$



Norm In $\mathbb{R}^3$

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2 + v_3^2} \tag{1.3}$$



Definition 1.2.4 (Definition 4)**.** If $\vec{v} = (v_1, v_2, \ldots, v_n)$ is a vector in $\mathbb{R}^n$, then the norm of v(called **length** or magnitude) is denoted by $||v||$ and defined by

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

1.2.2 Examples

Example 1

Given the following vector $\vec{v} = (-3, 2, 1)$ in $\mathbb{R}^3$, calculate the norm

$$||v|| = \sqrt{(-3)^2 + (2)^2 + (1)^2} = \sqrt{9 + 4 + 1} = \sqrt{14} \tag{1.4}$$

1.2.3 Unit Vectors

A vector with the norm valued at 1 is called a **unit vector**.

$$\vec{u} = \frac{1}{||\vec{v}||} \cdot \vec{v} \tag{1.5}$$

defines a unit vector that is in the same direction as $\vec{v}$.

Example: Normalizing A Vector

Find the unit vector $\vec{u}$ that has the same direction as $\vec{v} = (2, 2, -1)$.
The vector has length

$$||\vec{v}|| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$$
$$u = \frac{1}{3}(2, 2, -1) = (\frac{2}{3}, \frac{2}{3}, \frac{-1}{3})$$

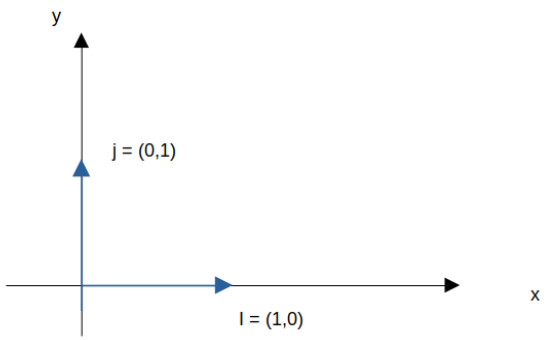
1.2.4 The Standard Unit Vectors

When a rectangular coordinate system is introduced in $\mathbb{R}^2$ or $\mathbb{R}$ he unit vectors in the positive directions of the coordinate axes are called the **standard unit vectors**.

In $\mathbb{R}^2$ these vectors are denoted by

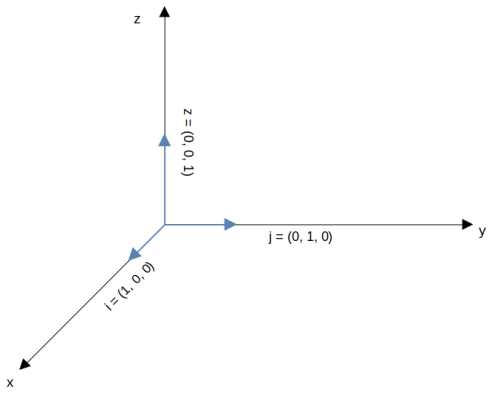
$$\vec{i} = (1, 0) \quad \text{and} \quad j = (0, 1) \tag{1.6}$$

and in $\mathbb{R}^3$ by



$$\vec{i} = (1,0,0) \quad \vec{j} = (0,1,0) \quad \vec{k} = (0,0,1)$$

Every vector $\vec{v} = (v_1, v_2)$ in $\mathbb{R}^2$ and every vector $\vec{v} = (v_1, v_2, v_3)$ in $\mathbb{R}^3$ can be



expressed as a linear combination of standard unit vectors by writing

$$\boldsymbol{v} = (v_1, v_2) = \boldsymbol{v_1}(1,0) + \boldsymbol{v_2}(0,1) = \boldsymbol{v_1} \cdot \vec{i} + \boldsymbol{v_2} \cdot \vec{j}$$
$$\boldsymbol{v} = (v_1, v_2, v_3) = \boldsymbol{v_1}(1,0,0) + \boldsymbol{v_2}(0,1,0) + v_3(0,0,1) = \boldsymbol{v_1} \cdot \vec{i} + \boldsymbol{v_2} \cdot \vec{j} + \boldsymbol{v_3} \cdot \vec{k}$$

Every vector $\vec{v} = (v_1, v_2, \ldots, v_n)$ in $\mathbb{R}^n$ can be expressed as

$$\vec{v} = (v_1, v_2, \ldots, v_n) = v_1\boldsymbol{e}_1 + v_2\boldsymbol{e_2} + \cdots + v_n\boldsymbol{e_n} \tag{1.7}$$

Example: Linear Combination of Standard Unit Vectors

$$(2, -3, 4) = 2\vec{i} - 3\vec{j} + 4\vec{k}$$
$$(7, 3, -4, 5) = 7\vec{e_1} + 3\vec{e_2} - 4\vec{e_3} + 5\vec{e_4}$$

1.2.5 Distance In $\mathbb{R}^n$

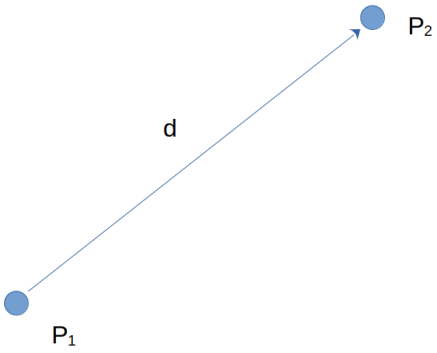
If P_1 and P_2 are points in $\mathbb{R}^2$ or $\mathbb{R}^3$ then the length of the vector $\vec{P_1P_2}$ is equal to the distance d between the two points.

Specifically, if $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are points in $\mathbb{R}^2$ then

$$d = ||\vec{P_1P_2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{1.8}$$

Similarly, the distance between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_3)$ in 3-space is

$$d(\vec{u}, \vec{v}) = ||\vec{P_1P_2}|| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \tag{1.9}$$



Definition 1.2.5 (Definition 5)**.** If $\vec{u} = (u_1, u_2, \ldots, u_n)$ and $\vec{v} = (v_1, v_2, \ldots, v_n)$ are points in $\mathbb{R}^n$, then we denote the distance between $\vec{u}$ and $\vec{v}$ by d$(\vec{u}, \vec{v})$ and define it to be

$$d(\vec{u}, \vec{v}) = ||\vec{u} - \vec{v}|| = \sqrt{(\vec{u_1} - \vec{v_1})^2 + (\vec{u_2} - \vec{v_2})^2 + \cdots + (\vec{u_n} - \vec{v_n})^2}$$

Example: Calculating distance in R^n

Given $\vec{u} = (1, 3, -2, 7)$, $\vec{v} = (0, 7, 2, 2)$, the distance is

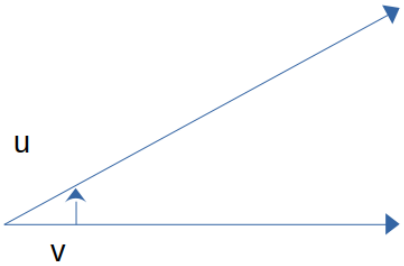
$$\begin{aligned} d(\vec{u}, \vec{v}) &= \sqrt{(1-0)^2 + (3-7)^2 + (-2-2)^2 + (7-2)^2} \\ &= \sqrt{58} \end{aligned}$$

1.3 Inner Products

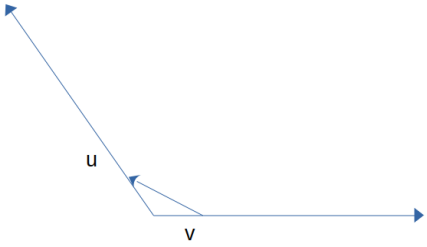
1.3.1 Dot Product

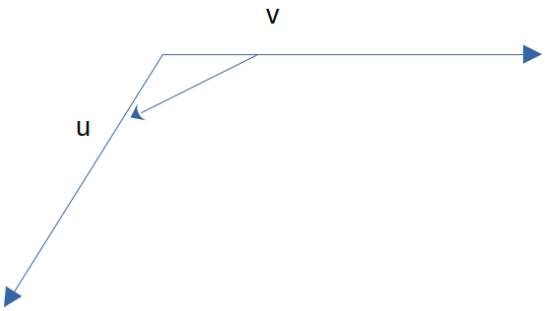
Let $\vec{u}$ and $\vec{v}$ be non-zero vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ that have been positioned so that their initial points coincide. We define the angle between $\vec{u}$ and $\vec{v}$ to be the angle θ determined by $\vec{u}$ and $\vec{v}$ that satisfies the inequalities.

$$0 \leq \theta \leq \pi$$



Definition 1.3.1 (Definition 5)**.** If $\vec{u}$ and $\vec{v}$ are non-zero vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$, and if θ is the angle between $\vec{u}$ and $\vec{v}$, then the dot product of $\vec{u}$ and $\vec{v}$ is denoted $\vec{u} \cdot \vec{v}$ and





is defined as

$$\vec{u} \cdot \vec{v} = ||\vec{u}||||\vec{v}|| \cos\theta \tag{1.10}$$

If $\vec{u} = \vec{0}$ or $\vec{u} = \vec{0}$, then we define $\vec{u} \cdot \vec{v}$ to be 0.

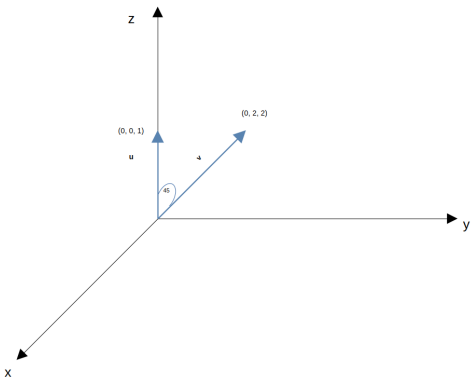
The sign of the dot product reveals information about the angle θ so that

$$cos\theta = \frac{\vec{u} \cdot \vec{v}}{||\vec{u}|| \cdot ||\vec{v}||} \tag{1.11}$$

Since $0 \leq \theta \leq \pi$, it follows that

- θ is acute if $\vec{u} - \vec{v} > 0$
- θ is obtense if $\vec{u} \cdot \vec{v} < 0$
- $\theta = \frac{\pi}{2}$ if $\vec{u} \cdot \vec{v} = 0$

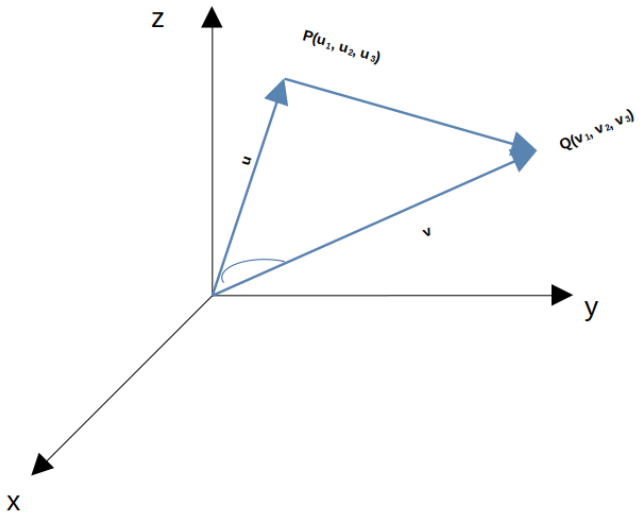
Example: Find The Dot Product



$$||\vec{u}|| = \sqrt{0+0+1} = 1$$
$$||\vec{v}|| = \sqrt{0+2^2+2^2} = \sqrt{8} = 2\sqrt{2}$$
$$cos\theta = cos(45^\circ) = \frac{\sqrt{2}}{2}$$
$$\vec{u}\cdot\vec{v} = ||\vec{u}||||\vec{v}||cos\theta = (1)(2\sqrt{2})(\frac{\sqrt{2}}{2})$$
$$= 2$$

1.3.2 Component Form of Dot Product

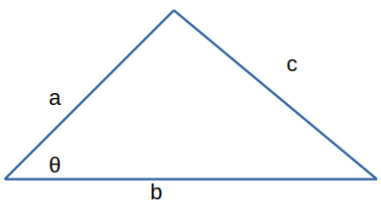
For computational purposes, it is desirable to have a formula that expresses dot product of two vectors in terms of components Let $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$



be two non-zero vecctors.
If θ is the angle between $\vec{u}$ and $\vec{v}$, then law of cosine yields

$$||\vec{PQ}||^2 = ||\vec{u}||^2 + ||\vec{v}||^2 - 2||\vec{u}||||\vec{v}||cos\theta \tag{1.12}$$

Recall the law of cosines



$$c = \sqrt{a^2 + b^2 - 2ab \cdot cos\theta} \quad \text{From law of cosines, relate to } \vec{PQ}$$

$$\text{Since } \vec{PQ} = \vec{v} - \vec{u}, \text{ we have } ||\vec{PQ}||^2 = ||\vec{v} - \vec{u}||^2$$

$$||\vec{PQ}||^2 = ||\vec{u}||^2 + ||\vec{v}||^2 - 2||\vec{u}||||\vec{v}||cos\theta$$

$$2||\vec{u}||||\vec{v}||cos\theta = ||\vec{u}||^2 + ||\vec{v}||^2 - ||\vec{PQ}||^2$$

$$||\vec{u}||||\vec{v}||cos\theta = \frac{1}{2}(||\vec{u}||^2 + ||\vec{v}||^2 - ||\vec{v} - \vec{u}||^2)$$

$$\vec{u} \cdot \vec{v} = \frac{1}{2}(||\vec{u}||^2 + ||\vec{v}||^2 - ||\vec{v} - \vec{u}||^2)$$

Consider the vectors $\vec{u}$, and $\vec{v}$

$$||\vec{u}||^2 = u_1^2 + u_2^2 + u_3^2$$

$$||\vec{v}||^2 = v_1^2 + v_2^2 + v_3^2$$

$$||\vec{v} - \vec{u}||^2 = (v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2$$

$$\because \vec{u} \cdot \vec{v} = \frac{1}{2}||\vec{u}||^2 + \frac{1}{2}||\vec{v}||^2 - \frac{1}{2}||\vec{v} - \vec{u}||^2$$

$$= \frac{1}{2}(u_1^2 + u_2^2 + u_3^2) + \frac{1}{2}(v_1^2 + v_2^2 + v_3^2) - \frac{1}{2}[(v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2]$$

$$= \frac{1}{2}(u_1^2 + u_2^2 + u_3^2) + \frac{1}{2}(v_1^2 + v_2^2 + v_3^2) - \frac{1}{2}$$

$$[(v_1^2 - 2v_1u_1 + u_1^2) + (v_2^2 - 2v_2u_2 + u_2^2) + (v_3^2 - 2v_3u_3 + u_3^2)]$$

$$= \frac{1}{2}[\cancel{u_1^2} + \cancel{u_2^2} + \cancel{u_3^2} + \cancel{v_1^2} + \cancel{v_2^2} + \cancel{v_3^2} - \cancel{v_1^2} + 2v_1u_1 - \cancel{u_1^2} - \cancel{v_2^2} + 2v_2u_2 - \cancel{u_2^2} - \cancel{v_3^2} + 2v_3u_3 - \cancel{u_3^2}]$$

$$= \frac{1}{2}[2v_1u_1 + 2v_2u_2 + 2v_3u_3]$$

$$= u_1v_1 + u_2v_2 + u_3v_3$$

$$= \vec{u}^T\vec{v}$$

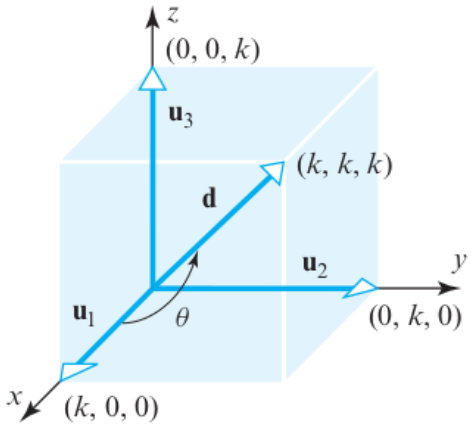
Definition 1.3.2 (Definition 6)**.** If $\vec{u} = (u_1, u_2, \ldots, u_n)$ and $\vec{v} = (v_1, v_2, \ldots, v_n)$ are vectors in $\mathbb{R}^n$, then dot product of $\vec{u}$ and $\vec{v}$ is denoted by $\vec{u} \cdot \vec{v}$ and defined by

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n \tag{1.13}$$

Example: Geometry Problem Solved Using Dot Product

Find the angle between a diagonal of a cube and one of its edges Solution Let k be the length of an edge and introduce a coordinate system as shown in Figure 3.2.7. If we let $\mathbf{u}_1 = (k, 0, 0)$, $\mathbf{u}_2 = (0, k, 0)$, and $\mathbf{u}_3 = (0, 0, k)$, then the vector

$$\mathbf{d} = (k, k, k) = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$$



is a diagonal of the cube. It follows from Formula (13) that the angle θ between $\mathbf{d}$ and the edge $\mathbf{u}_1$ satisfies

$$\cos\theta = \frac{\mathbf{u}_1 \cdot \mathbf{d}}{\|\mathbf{u}_1\|\,\|\mathbf{d}\|} = \frac{k^2}{(k)\left(\sqrt{3k^2}\right)} = \frac{1}{\sqrt{3}}$$

With the help of a calculator we obtain

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 54.74^\circ$$

In the special case where $\mathbf{u} = \mathbf{v}$ in Definition 4 , we obtain the relationship

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2 = \|\mathbf{v}\|^2$$

1.3.3 Algebraic Properties of The Dot Product

This yields the following formula for expressing the length of a vector in terms of a dot product:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

THEOREM 3.2.2 If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in R^n, and if k is a scalar, then:
(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ [Symmetry property]
(b) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ [Distributive property]
(c) $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$ [Homogeneity property]
(d) $\mathbf{v} \cdot \mathbf{v} \geq 0$ and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$ [Positivity property]

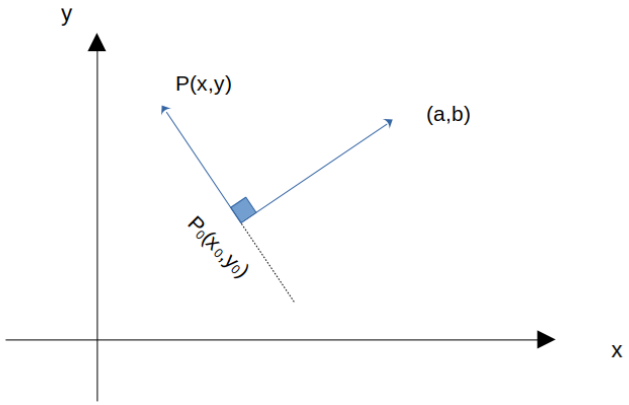
Calculating With Dot Products

$$\begin{aligned}(\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} + 4\mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + 4\mathbf{v}) - 2\mathbf{v} \cdot (3\mathbf{u} + 4\mathbf{v}) \\ &= 3(\mathbf{u} \cdot \mathbf{u}) + 4(\mathbf{u} \cdot \mathbf{v}) - 6(\mathbf{v} \cdot \mathbf{u}) - 8(\mathbf{v} \cdot \mathbf{v}) \\ &= 3\|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) - 8\|\mathbf{v}\|^2\end{aligned}$$

Orthogonality

Definition 1.3.3 (Definition 7)**.** Two non-zero vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ are orthogonal if $\vec{u} \cdot \vec{v} = 0$

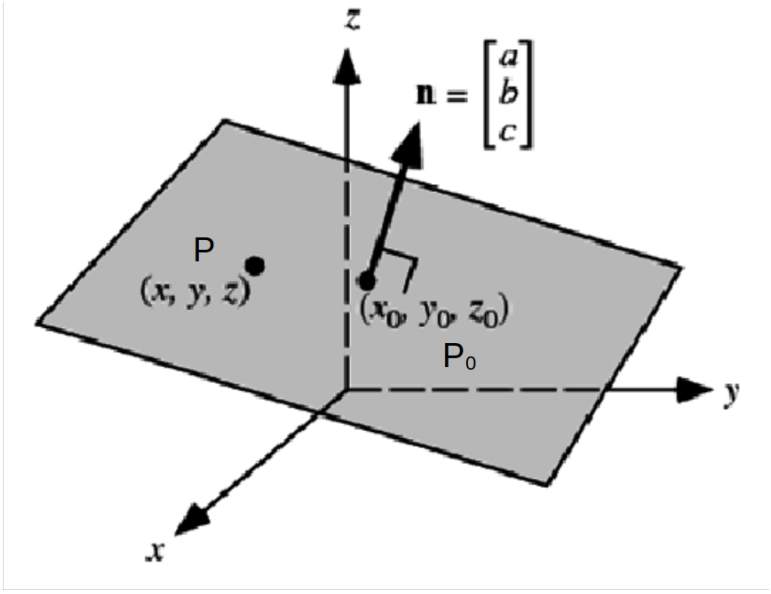
1.3.4 Lines and Planes Determined By Points and Normals



Line through point $P_0(x_0,y_0,z_0)$ with normal n = (a,b)

2D

Plane through Point $P_0(x_0, y_0, z_0)$ with normal n



3D

One way of specifying slope is to use non-zero vector $\vec{n}$, called normal that is orthogonal to line or place in question.
Both the line and plane are represented by the vector equation

$$\vec{n} \cdot \vec{P_0 P} = 0 \tag{1.14}$$

where P is either an arbitary point (x, y) on the line or an arbitrary point (x, y, z) in the plane.

The vector $\vec{P_0P}$ can be expressed in terms of components as

$$\vec{P_0P} = (x - x_0, y - y_0) \quad \text{line}$$
$$\vec{P_0P} = (x - x_0, y - y_0, z - z_0) \quad \text{plane}$$

Thus we have $\vec{n} \cdot \vec{P_0P}$ as:

$$\begin{bmatrix} a \\ b \end{bmatrix} (x - x_0, y - y_0) = a(x - x_0) + b(y - y_0) = 0 \quad \text{[line]} \tag{1.15}$$

and

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} (x - x_0, y - y_0, z - z_0) = a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \quad \text{[plane]} \tag{1.16}$$

These are called **point normal equations**

Example: Point Normal Equations

You have a point (3,-7) with normal $\vec{n} = (6, 1)$ thus you have

$$n \cdot \vec{P_0P} = a(x - x_0) + b(y - y_0) = 0$$
$$\implies 6(x - 3) + (y + 7) = 0$$

Example: Point Normal Equations In 3D

You have a point (3,0, 7) with normal $\vec{n} = (4, 2, 5)$ thus you have

$$n \cdot \vec{P_0P} = a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$
$$\implies 4(x - 3) + 2y + 5(z - 7) = 0$$

THEOREM 3.3.1
(a) If a and b are constants that are not both zero, then an equation of the form

$$ax + by + c = 0$$

represents a line in R^2 with normal $\mathbf{n} = (a, b)$.
(b) If a, b, and c are constants that are not all zero, then an equation of the form

$$ax + by + cz + d = 0$$

represents a plane in R^3 with normal $\mathbf{n} = (a, b, c)$.

Example: Vectors Orthogonal To Lines and Planes Through Origin

(a) The equation $ax + by = 0$ represents a line through the origin in R^2. Show that the vector $\mathbf{n}_1 = (a, b)$ formed from the coefficients of the equation is orthogonal to the line, that is, orthogonal to every vector along the line.
(b) The equation $ax + by + cz = 0$ represents a plane through the origin in R^3. Show that the vector $\mathbf{n}_2 = (a, b, c)$ formed from the coefficients of the equation is orthogonal to the plane, that is, orthogonal to every vector that lies in the plane.

$$\text{Given } ax + by = 0$$

$$\vec{n_1} \cdot \vec{P_0P} = (a, b)^T \cdot (x - x_0, y - y_0) = 0$$

where $x_0 = 0$ and $y_0 = 0$, thus

$$\vec{n_1} \cdot \vec{P_0P} = (a, b)^T \cdot (x, y) = 0$$
$$\text{and} \vec{n_2} \cdot \vec{P_0P} = (a, b, c)^T \cdot (x, y, z) = 0$$

such that

$$ax + by = 0 \quad \text{and } ax + by + c = 0$$

are called **homogenous equations**

1.3.5 Decomposing A Vector Into A Sum of Two Orthogonal Vectors

In many applications it is necessary to "decompose" a vector u into a sum of two terms, one term being a scalar multiple of a specified nonzero vector a and the other term being orthogonal to a. For example, if $\vec{u}$ and $\vec{a}$ are vectors in $\mathbb{R}^2$ that are positioned so their initial points coincide at a point Q, then we can create such a decomposition as follows:

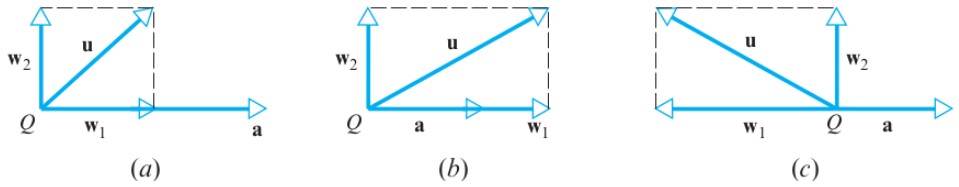
- Drop a perpendicular from the tip of $\vec{u}$ to the line through $\vec{a}$
- Construct the vector w_1 from Q to the foot of the perpendicular
- Construct the vector $w_2 = \vec{u} - \vec{w_1}$

The vectors $\vec{w_1}$ and $\vec{w_2}$ in the **projection theorem** have associated names

1. $\vec{w_1}$:
    - Orthogonal projection $\vec{u}$ on $\vec{a}$
    - The vector component of $\vec{u}$ along $\vec{a}$

- Otherwise denoted as $proj_{\vec{a}}\vec{u}$

2. $\vec{w}_2$:
   - Vector component of $\vec{u}$ orthogonal to $\vec{a}$
   - $\vec{u} - proj_a\vec{u}$



▲ Figure 3.3.2 Three possible cases.

Theorem 1 (Projection Theorem)**.** *If* $\mathbf{u}$ *and* $\mathbf{a}$ *are vectors in* R^n*, and if* $\mathbf{a} \neq 0$*, then* $\mathbf{u}$ *can be expressed in exactly one way in the form* $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$*, where* $\mathbf{w}_1$ *is a scalar multiple of* $\mathbf{a}$ *and* $\mathbf{w}_2$ *is orthogonal to* $\mathbf{a}$*.*

Proof:
Since the vector $\mathbf{w}_1$ is to be a scalar multiple of $\mathbf{a}$, it must have the form

$$\mathbf{w}_1 = k\mathbf{a}$$

Our goal is to find a value of the scalar k and a vector $\mathbf{w}_2$ that is orthogonal to a such that

$$\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$$

We can determine k by using (7) to rewrite (8) as

$$\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2 = k\mathbf{a} + \mathbf{w}_2$$

and then applying Theorems 3.2.2 and 3.2.3 to obtain

$$\mathbf{u} \cdot \mathbf{a} = (k\mathbf{a} + \mathbf{w}_2) \cdot \mathbf{a} = k\|\mathbf{a}\|^2 + (\mathbf{w}_2 \cdot \mathbf{a})$$

Since $\mathbf{w}_2$ is to be orthogonal to $\mathbf{a}$, the last term in (9) must be 0 , and hence k must satisfy the equation

$$\mathbf{u} \cdot \mathbf{a} = k\|\mathbf{a}\|^2$$

from which we obtain

$$k = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}$$

as the only possible value for k. The proof can be completed by rewriting (8) as

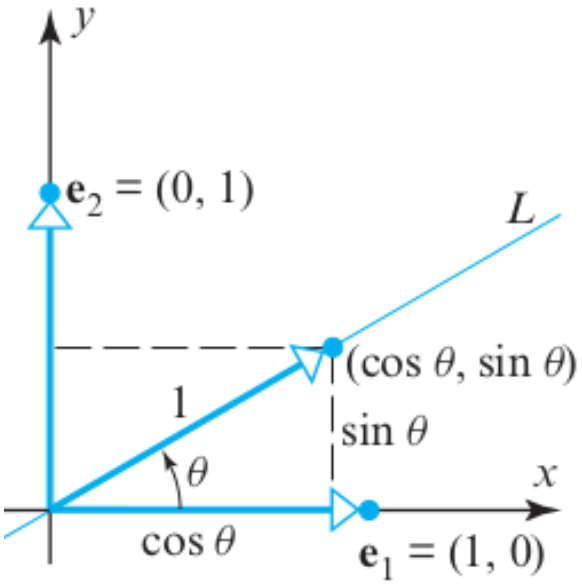
$$\mathbf{w}_2 = \mathbf{u} - \mathbf{w}_1 = \mathbf{u} - k\mathbf{a} = \mathbf{u} - \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}$$

In summary,

$$\text{proj}_a\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} \quad \text{(vector component of } \mathbf{u} \text{ along a)}$$

$$\mathbf{u} - \text{proj}_a\,\mathbf{u} = \mathbf{u} - \frac{\mathbf{u}\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} \text{ (vector component of } \mathbf{u} \text{ orthogonal to a)}$$

Example: Orthogonal Projection on a Line



Find the orthogonal projections of the vectors $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$ on the line L that makes an angle θ with the positive x-axis in R^2.

Solution As illustrated in Figure 3.3.3, $\mathbf{a} = (\cos\theta, \sin\theta)$ is a unit vector along the line L, so our first problem is to find the orthogonal projection of $\mathbf{e}_1$ along $\mathbf{a}$. Since

$$\|\mathbf{a}\| = \sqrt{\sin^2\theta + \cos^2\theta} = 1 \quad \text{and} \quad \mathbf{e}_1\cdot\mathbf{a} = (1,0)\cdot(\cos\theta,\sin\theta) = \cos\theta$$

it follows from Formula (10) that this projection is

$$\text{proj}_{\mathbf{a}}\,\mathbf{e}_1 = \frac{\mathbf{e}_1\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} = (\cos\theta)(\cos\theta,\sin\theta) = \left(\cos^2\theta, \sin\theta\cos\theta\right)$$

Similarly, since $\mathbf{e}_2\cdot\mathbf{a} = (0,1)\cdot(\cos\theta,\sin\theta) = \sin\theta$, it follows from Formula (10) that

$$\text{proj}_{\mathbf{a}}\,\mathbf{e}_2 = \frac{\mathbf{e}_2\cdot\mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} = (\sin\theta)(\cos\theta,\sin\theta) = \left(\sin\theta\cos\theta, \sin^2\theta\right)$$

Example: Vector Component of u Along a

Let $\mathbf{u} = (2, -1, 3)$ and $\mathbf{a} = (4, -1, 2)$. Find the vector component of $\mathbf{u}$ along a and the vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$. Solution

$$\mathbf{u} \cdot \mathbf{a} = (2)(4) + (-1)(-1) + (3)(2) = 15$$
$$\|\mathbf{a}\|^2 = 4^2 + (-1)^2 + 2^2 = 21$$

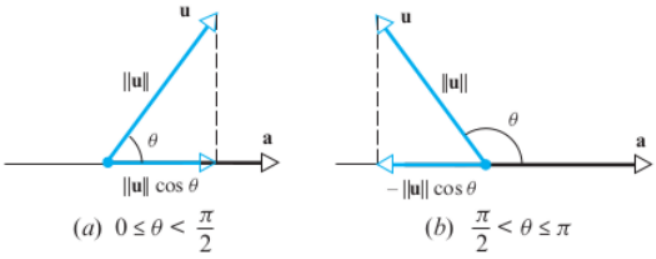
Thus the vector component of $\mathbf{u}$ along $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}}\,\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a} = \frac{15}{21}(4, -1, 2) = \left(\frac{20}{7}, -\frac{5}{7}, \frac{10}{7}\right)$$

and the vector component of $\mathbf{u}$ orthogonal to $\mathbf{a}$ is

$$\mathbf{u} - \text{proj}_{\mathbf{a}}\,\mathbf{u} = (2, -1, 3) - \left(\frac{20}{7}, -\frac{5}{7}, \frac{10}{7}\right) = \left(-\frac{6}{7}, -\frac{2}{7}, \frac{11}{7}\right)$$

1.3.6 Defining Projections



(a) $0 \le \theta < \frac{\pi}{2}$ (b) $\frac{\pi}{2} < \theta \le \pi$

Sometimes we will be more interested in the norm of the vector component of $\mathbf{u}$ along a than in the vector component itself. A formula for this norm can be derived as follows:

$$\|\text{proj}_{\mathbf{a}}\,\mathbf{u}\| = \left\|\frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\mathbf{a}\right\| = \left|\frac{\mathbf{u} \cdot \mathbf{a}}{\|\mathbf{a}\|^2}\right| \|\mathbf{a}\| = \frac{|\mathbf{u} \cdot \mathbf{a}|}{\|\mathbf{a}\|^2}\|\mathbf{a}\|$$

where the second equality follows from part (c) of Theorem 3.2.1 and the third from the fact that $\|\mathbf{a}\|^2 > 0$. Thus,

$$\|\text{proj}_{\mathbf{a}}\,\mathbf{u}\| = \frac{|\mathbf{u} \cdot \mathbf{a}|}{\|\mathbf{a}\|}$$

If θ denotes the angle between $\mathbf{u}$ and $\mathbf{a}$, then $\mathbf{u} \cdot \mathbf{a} = \|\mathbf{u}\|\|\mathbf{a}\| \cos\theta$, so (12) can also be written as

$$\|\text{proj}_{\mathbf{a}}\,\mathbf{u}\| = \|\mathbf{u}\||\cos\theta|$$

1.3.7 Distance In $\mathbb{R}^n$

We will now show how orthogonal projections can be used to solve the following three distance problems:

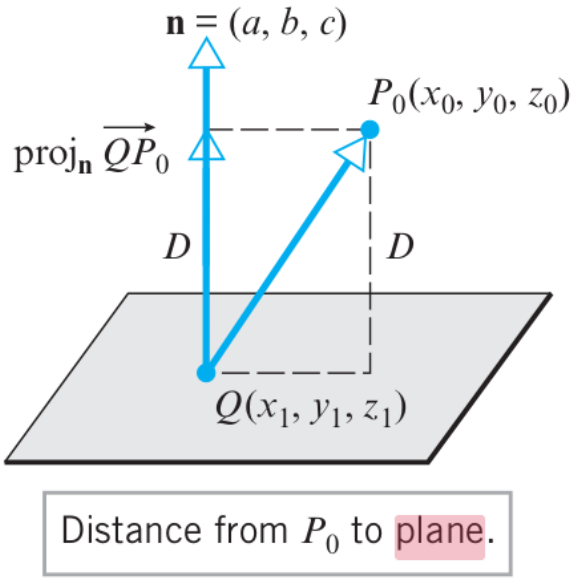
- **Problem 1**. Find the distance between a point and a line in R^2.
- **Problem 2**. Find the distance between a point and a plane in R^3.
- **Problem 3**. Find the distance between two parallel planes in R^3.

Theorem 2 (Distance Theorem). *a) In* $\mathbb{R}^2$ *the distance D between the point* $P_0(x_0, y_0)$ *and the line* $ax + by + c = 0$ *is*

$$D = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

b) In $\mathbb{R}^3$ *the distance D between the point* $P_0(x_0, y_0, z_0)$ *and the plane* $ax + by + cz + d = 0$ *is*

$$D = \frac{|ax_0 + by_0 + cz + d|}{\sqrt{a^2 + b^2 + c^2}}$$



(b) The underlying idea of the proof is illustrated in Figure 3.3.6. As shown in that figure, let $Q\,(x_1, y_1, z_1)$ be any point in the plane, and let $\mathbf{n} = (a, b, c)$ be a normal vector to the plane that is positioned with its initial point at Q. It is now evident that the distance D between P_0 and the plane is simply the length (or norm) of the orthogonal projection of the vector $\overrightarrow{QP_0}$ on $\mathbf{n}$, which by Formula (12) is

$$D = \left\|\text{proj}_{\mathbf{n}}\, \overrightarrow{QP_0}\right\| = \frac{\left|\overrightarrow{QP_0} \cdot \mathbf{n}\right|}{\|\mathbf{n}\|}$$

But

$$\overrightarrow{QP_0} = (x_0 - x_1, y_0 - y_1, z_0 - z_1)$$
$$\overrightarrow{QP_0} \cdot \mathbf{n} = a\,(x_0 - x_1) + b\,(y_0 - y_1) + c\,(z_0 - z_1)$$
$$\|\mathbf{n}\| = \sqrt{a^2 + b^2 + c^2}$$

Thus

$$D = \frac{|a\,(x_0 - x_1) + b\,(y_0 - y_1) + c\,(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}}$$

Since the point $Q\,(x_1, y_1, z_1)$ lies in the given plane, its coordinates satisfy the equation of that plane; thus

$$ax_1 + by_1 + cz_1 + d = 0$$

or

$$d = -ax_1 - by_1 - cz_1$$

$$a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1) = ax_0 - ax_1 + by_0 - by_1 + cz_0 - cz_1$$

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Example: Distance Between A Point And A Plane

Find the distance D between the point $(1, -4, -3)$ and the plane $2x - 3y + 6z = -1$.

Solution

Since the distance formulas in Theorem 3.3.4 require that the equations of the line and plane be written with zero on the right side, we first need to rewrite the equation of the plane as

$$2x - 3y + 6z + 1 = 0$$

from which we obtain

$$D = \frac{|2(1) + (-3)(-4) + 6(-3) + 1|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{|-3|}{7} = \frac{3}{7}$$

Example: Distance Between A Point And A Plane (2)

Find the distance from point $Q(3, 8, -1)$ to the plane $4(x + 2) - (y + 4) + 2z = 0$

$$\text{Distance} = \frac{\vec{PQ} \cdot \vec{n}}{||\vec{n}||}$$

P is the origin point that makes the hyperplane zero

$$P(-2, -5, 0),\ \vec{n}(4, -1, 2) \text{ and } Q(3, 8, -1)$$

$$\begin{aligned}\vec{PQ} &= (3-(-2), 8-(-5), -1-0)\\ &= (5, 13, -1)\\ D &= \frac{|\vec{PQ}\cdot\vec{n}}{||\vec{n}||}\\ &= \frac{|20-13-2|}{\sqrt{4^2+(-1)^2+(2^2)}}\\ &= \frac{5}{\sqrt{21}}\end{aligned}$$

36

Example: Distance Between Parallel Planes

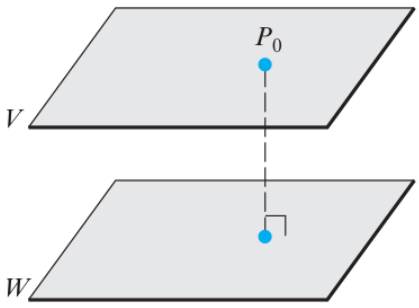
The planes

$$x + 2y - 2z = 3 \text{ and } 2x + 4y - 4z = 7$$

are parallel since their normals, $(1, 2, -2)$ and $(2, 4, -4)$, are parallel vectors. Find the distance between these planes.

Solution

To find the distance D between the planes, we can select an arbitrary point in one of the planes and compute its distance to the other plane. By setting $y = z = 0$ in the equation $x + 2y - 2z = 3$, we obtain the point $P_0(3, 0, 0)$ in this plane. From (16), the distance between P_0 and the plane $2x + 4y - 4z = 7$ is



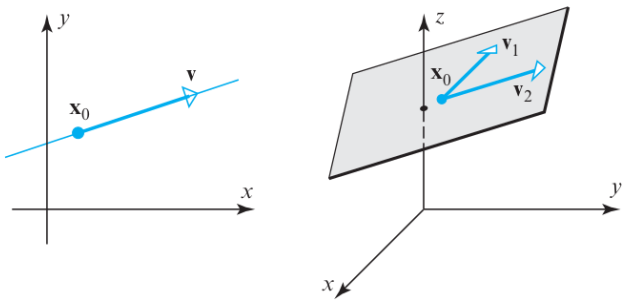
▲ Figure 3.3.7 The distance between the parallel planes V and W is equal to the distance between P_0 and W.

$$D = \frac{|2(3) + 4(0) + (-4)(0) - 7|}{\sqrt{2^2 + 4^2 + (-4)^2}} = \frac{1}{6}$$

Chapter 2

Geometry of Linear Systems

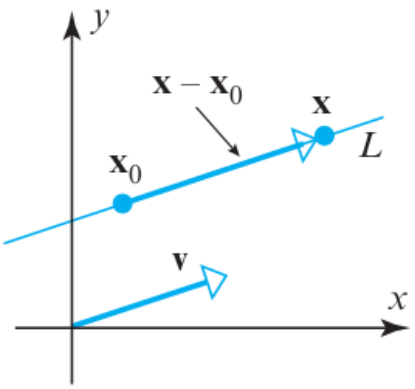
2.1 Linear Systems



Deriving an equation for the line L that contains the point $\vec{x}_0$ and is parallel to v.
If $\vec{x}$ is a general point on such a line, the vector $\vec{x} - \vec{x}_0$ will be some scalar multiple of $\vec{v}$, say

$$\vec{x} - \vec{x}_0 = t\vec{v} \text{ or } \vec{x} = \vec{x}_0 + t\vec{v}$$

As the variable t (called parameter) varies from ∞ to ∞



2.1.1 Parameterization

Theorem 3 (Parameterization). *Let L be the line in $\mathbb{R}^2$ or $\mathbb{R}^3$ that contains the point x_0 and is parallel to the nonzero vector v. Then the equation of the line through x_0 that is parallel to v is*

$$x = x_0 + tv$$

if $x_0 = 0$, then the line passes through the origin and the equation has the form

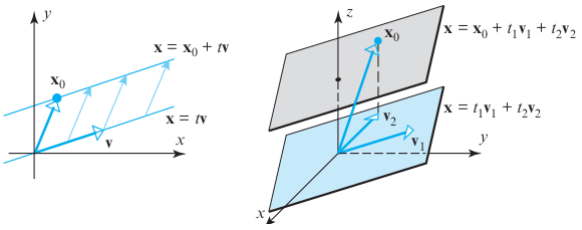
$$x = tv$$

Theorem 4 (Parameterization In $\mathbb{R}^3$). *Let W be the plane in $\mathbb{R}^3$ that contains the point x_0 and is parallel to the noncollinear vectors v_1 and v_2 . Then an equation of the plane through x_0 that is parallel to v_1 and v_2 is given by*

$$x = x_0 + t_1v_1 + t_2v_2$$

If $x_0 = 0$, then the plane passes through the origin and the equation has the form

$$x = t_1v_1 + t_2v_2$$



Below, we will define what are called **vector forms** of a line and plan in $\mathbb{R}^n$.

Definition 2.1.1 (Definition 1). If x_0 and v are vectors in $\mathbb{R}^n$, and if v is nonzero, then the equation

$$x = x_0 + tv \tag{2.1}$$

defines the **line through x_0 that is parallel to** v. In the special case where $x_0 = 0$, the line is said to **pass through the origin**

Definition 2.1.2 (Definition 2). If $x_0, v_1 textand v_2$ are vectors in $\mathbb{R}^n$, and if v_1 and v_2 are not collinear then the equation

$$x = x_0 + t_1v_1 + t_2v_2 \tag{2.2}$$

defines the **plane through x_0 that is parallel to** v_1 and v_2. In the special case where $x_0 = 0$, the plane is said to **pass through the origin**

Example: Vector and Parametric Equations of Lines in $\mathbb{R}^2$ and $\mathbb{R}^3$

- Find a vector equation and parametric equations of the line in $\mathbb{R}^2$ that passes through the origin and is parallel to the vector v = (-2, 3)
- Find a vector equation and parametric equations of the line in $\mathbb{R}^3$ that passes through the point P_0 (1, 2, -3) and is parallel to the vector $v = (4, -5, 1)$
- Use the vector equation obtained in part b to find two points on the line that are different from P_0

Solution (a)
It follows from (5) with $\mathbf{x}_0 = \mathbf{0}$ that a vector equation of the line is $\mathbf{x} = t\mathbf{v}$. If we let $\mathbf{x} = (x, y)$, then this equation can be expressed in vector form as

$$(x, y) = t(-2, 3)$$

Equating corresponding components on the two sides of this equation yields the parametric equations

$$x = -2t, \quad y = 3t$$

Solution (b)
It follows from (5) that a vector equation of the line is $\mathbf{x} = \mathbf{x}_0 + t\mathbf{v}$. If we let $\mathbf{x} = (x, y, z)$, and if we take $\mathbf{x}_0 = (1, 2, -3)$, then this equation can be expressed in vector form as

$$(x, y, z) = (1, 2, -3) + t(4, -5, 1)$$

Equating corresponding components on the two sides of this equation yields the parametric equations

$$x = 1 + 4t, \quad y = 2 - 5t, \quad z = -3 + t$$

Solution (c)
A point on the line represented by Equation (7) can be obtained by substituting a specific numerical value for the parameter t. However, since $t = 0$ produces $(x, y, z) = (1, 2, -3)$, which is the point P_0, this value of t does not serve our purpose. Taking $t = 1$ produces the point $(5, -3, -2)$ and taking $t = -1$ produces the point $(-3, 7, -4)$. Any other distinct values for t (except $t = 0$) would work just as well.

Example: Vector and Parametric Equations of a Plane in $\mathbb{R}^3$

Find vector and parametric equations of the plane $x - y + 2z = 5$.
Solution
We will find the parametric equations first. We can do this by solving the equation for any one of the variables in terms of the other two and then using those two

variables as parameters. For example, solving for x in terms of y and z yields

$$x = 5 + y - 2z$$

and then using y and z as parameters t_1 and t_2, respectively, yields the parametric equations

$$x = 5 + t_1 - 2t_2, \quad y = t_1, \quad z = t_2$$

To obtain a vector equation of the plane we rewrite these parametric equations as

$$(x, y, z) = (5 + t_1 - 2t_2, t_1, t_2)$$

or, equivalently, as

$$(x, y, z) = (5, 0, 0) + t_1(1, 1, 0) + t_2(-2, 0, 1)$$

Example: Vector and Parametric Equations of Lines and Planes in $\mathbb{R}^4$

- Find vector and parametric equations of the line through the origin of $\mathbb{R}^4$ that is parallel to the vector $v = (5, -3, 6, 1)$.
- Find vector and parametric equations of the plane in $\mathbb{R}^4$ that passes through the point $\boldsymbol{x0} = (2, -1, 0, 3)$ and is parallel to both $\boldsymbol{v_1} = (1, 5, 2, -4)$ and $\boldsymbol{v_2} = (0, 7, -8, 6)$.

Solution (a)
If we let $\mathbf{x} = (x_1, x_2, x_3, x_4)$, then the vector equation $\mathbf{x} = t\mathbf{v}$ can be expressed as

$$(x_1, x_2, x_3, x_4) = t(5, -3, 6, 1)$$

Equating corresponding components yields the parametric equations

$$x_1 = 5t, \quad x_2 = -3t, \quad x_3 = 6t, \quad x_4 = t$$

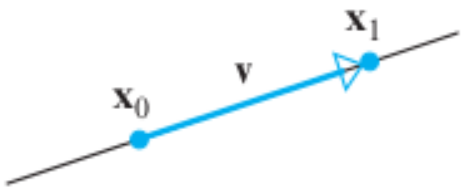
Solution (b)
The vector equation $\mathbf{x} = \mathbf{x}_0 + t_1\mathbf{v}_1 + t_2\mathbf{v}_2$ can be expressed as

$$(x_1, x_2, x_3, x_4) = (2, -1, 0, 3) + t_1(1, 5, 2, -4) + t_2(0, 7, -8, 6)$$

which yields the parametric equations

$$\begin{aligned} x_1 &= 2 + t_1 \\ x_2 &= -1 + 5t_1 + 7t_2 \\ x_3 &= 2t_1 - 8t_2 \\ x_4 &= 3 - 4t_1 + 6t_2 \end{aligned}$$

2.1.2 Lines Through Two Points In $\mathbb{R}^n$



If $\mathbf{x}_0$ and $\mathbf{x}_1$ are distinct points in R^n, then the line determined by these points is parallel to the vector $\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$ (Figure 3.4.5), so it follows from (5) that the line can be expressed in vector form as

$$\mathbf{x} = \mathbf{x}_0 + t\,(\mathbf{x}_1 - \mathbf{x}_0)$$

or, equivalently, as

$$\mathbf{x} = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$$

These are called the **two-point vector equations** of a line in R^n.

Example:A Line Through Two Points in $\mathbb{R}^2$

Find vector and parametric equations for the line in $\mathbb{R}^2$ that passes through the points $P(0,7)$ and $Q(5,0)$.

Solution

It does not matter,which order we take o be $\boldsymbol{x_1}$, we will choose $\boldsymbol{x_0} = (0,7)$ and $\boldsymbol{x_1} = (5,0)$.

It follows that

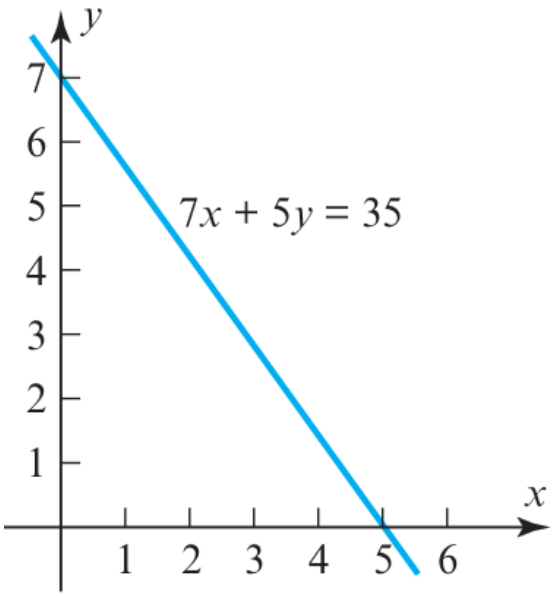
$$\boldsymbol{x_1} - \boldsymbol{x_0} = (5,-7)$$

hence

$$(x,y) = (0,7) + t(5,-7)$$
$$x = 5t \quad , y = 7 - 7t$$

Eliminating Parameter t

$$t = \frac{x}{5}, \quad y = 7 - \frac{7}{5}x$$
$$5y = 35 - 7x$$
$$7x + 5y = 35$$



2.1.3 Line Segment From One Point To Another

Definition 2.1.3 (Definition 3)**.** If $\mathbf{x}_0$ and $\mathbf{x}_1$ are vectors in R^n, then the equation

$$\mathbf{x} = \mathbf{x}_0 + t\,(\mathbf{x}_1 - \mathbf{x}_0) \quad (0 \leq t \leq 1)$$

defines the line segment from $\mathbf{x_0}$ to $\mathbf{x_1}$. When convenient, Equation (13) can be written as

$$\mathbf{x} = (1-t)\mathbf{x}_0 + t\mathbf{x}_1 \quad (0 \leq t \leq 1)$$

Example: A Line Segment from One Point to Another in $\mathbb{R}^2$

It follows that the line segment in R^2 from $\mathbf{x}_0 = (1,-3)$ to $\mathbf{x}_1 = (5,6)$ can be represented either by the equation

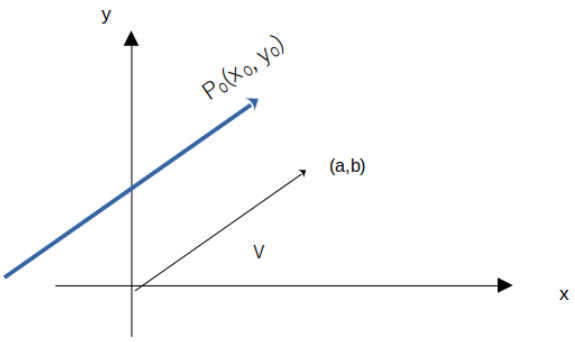
$$\mathbf{x} = (1,-3) + t(4,9) \quad (0 \leq t \leq 1)$$

or by the equation

$$\mathbf{x} = (1-t)(1,-3) + t(5,6) \quad (0 \leq t \leq 1)$$

2.1.4 Parametric Equations of Lines

A line in 2D or 3D space can be determined uniquely by specifying a point on that line and a **nonzero** vector parallel to the line.
Consider a line L in 3-space that passes through the point $P_0(x_0, y_0, z_0)$ and parallel to the nonzero vector $\vec{v} = (a,b,c)$. Then L consists precisely of those points $P(x,y,z)$ for which the vector $\vec{P_0P}$ is parallel to $\vec{v}$.



In other words, the point $P(x, y, z)$ is on L if and only if $\vec{P_0P}$ is a scalar multiple of $\vec{v}$.

$$\vec{P_0P} = t\vec{v}$$

The equation can be written as

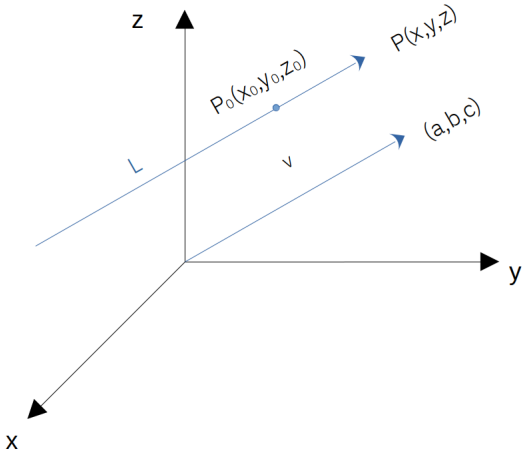
$$(x - x_0, y - y_0, z - z_0) = (t_a, t_b, t_c) \tag{2.3}$$

which implies

$$x - x_0 = t_a \quad y - y_0 = t_b \quad z - z_0 = t_c$$

Thus, L can be described by the parametric equation

$$x = x_0 + at \quad y = y_0 + bt \quad cz = z - z_0$$



Example: Find parametric equations of the line

Find the parametric equation of the line passing through (4,2) and parallel to $v = (-1, 5)$

Solution:

$$x_0 = 4 \quad y_0 = 2$$
$$a = -1 \quad b = 5$$
$$\therefore x = 4 - t \quad y = 2 + 5t$$

Example: Find parametric equations of the line in $\mathbb{R}^3$

Find the parametric equations of the line L passing through points $P_1(2, 4, -1)$ and $P_2(5, 0, 7)$.
Solution:
The vector $vecP_1P_2 = P_2 - P_1 = (5 - 2, 0 - 4, 7 - (-1)) = (3, -4, 8)$ is parallel to L and the point $P_1(2, 4, -1)$ lies on L

$$x = 2 + 3t$$
$$y = 4 - 4t$$
$$z = -1 + 8t$$

Alternative Solution:
The vector $\vec{P_1P_2} = (3, -4, 8)$ is parallel to the line, so it can be used as v. For $\vec{r_0}$ or $\vec{x_0}$ we can use either the vector from the origin to P_1 or origin to P_2.

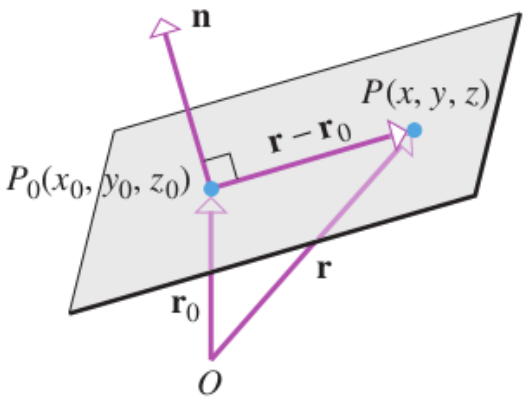
$$\vec{r_0} = (2, 4, -1)$$

Thus

$$\boldsymbol{r} = \vec{r_0} + t\vec{v}$$
$$(x, y, z) = (2, 4, -1) + t(3, -4, 8)$$

2.2 Planes In 3-Space

A uniqe plane is determined by a point in the plane and a nonzero vector normal to the plane. In contrast, a unique plane is not determined by a point in the plane and a nonzero vector parallel to the plane.
The unique plane is also determined by three non-collinear points that lie in the plane.



2.2.1 Determining Parallel Planes and Equations of Planes

Example: Determine Whether The Planes Are Parallel

Given

$$3x - 4y + 5 = 0 \quad -6x + 8y - 10z - 4 = 0$$

determine whether the planes are parallel

Solution:

Two planes are parallel if their normals are parallel

$$\vec{n_1} = (3, -4, 5) \quad \vec{n_2} = (-6, 8, -10)$$

Since $\vec{n_2}$ is a scalar multiple of $\vec{n_1}$, the normals are parallel

Example: Finding Equations of Plane Through Points and Normal

Find an equation of the plane through the point $(2, -5, 3)$ and the normal $(3, 6, 5)$

Solution:

Recall that we have

$$n = (a, b, c),\ \boldsymbol{r} = (x, y, z) \text{ and } \boldsymbol{r_0} = (x_0, y_0, z_0)$$

$$\boldsymbol{r} - \boldsymbol{r_0} = \vec{P_0P}$$

$$\vec{n} \cdot (\boldsymbol{r} - \boldsymbol{r_0}) = 0$$

$$(a, b, c) \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$3(x - 2) + 6(y - (-5)) + 5(z - 3) = 3x - 6 + 6y + 30 + 5z - 15 = 0$$

$$\implies 3x + 6y + 5z = -9$$

Example: Finding Equations of Plane Through Points

Find an equation of the plane through $\vec{P_1}(1,2,-1)$, $\vec{P_2}(2,3,1)$ and $\vec{P_3}(3,-1,2)$
Solution:
Since the points $\vec{P_1}$, $\vec{P_2}$, $\vec{P_3}$ lie in the plane, the vectors $\vec{P_1P_2} = (1,1,2)$ and $\vec{P_1P_3} = (2,-3,3)$ are parallel to the plane

$$\vec{P_1P_2} \times \vec{P_1P_3} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 2 \\ 2 & -3 & 3 \end{vmatrix} = 9\vec{i} + j - 5k = (9,1,-5)$$

is normal to the plane since it is orthogonal to both $\vec{P_1P_2}$ and $\vec{P_1P_3}$.
By using this normal and the point $P_1(1,2,-1)$ in the plane, we obtain

$$9(x-1) + (y-2) - 5(z+1) = 0$$
$$9x + y - 5z - 16 = 0$$

Example: Determining Whther Line Is Parallel To Plane

Determine whether the line

$$x = 3 + 8t, \quad y = 4 + 5t \quad z = -3 - t$$

is parallel to the plane $x - 3y + 5z = 12$
Solution:
The vector $\vec{v} = (8,5,-1)$, since $\vec{x} = \boldsymbol{x_0} + t\boldsymbol{v}$, is parallel to the line and the normal is (1,-3,5).
For the line and plane to be parallel, the vectors $\vec{v}$ and $\vec{n}$ must be orthogonal.

$$\begin{aligned} \vec{v}^T \cdot \vec{n} &= (8,5,-1)^T \cdot (1,-3,5) \\ &= 8 - 15 - 5 = 12 \quad \text{this is nonzero} \end{aligned}$$

Thus the line and plane are not parallel.

Example: Finding Intersection Between Line and Plane

Find the intersection of the line and plane given that the line is

$$x = 3 + 8t \quad y = 4 + 5t \quad z = -3 - t$$

and the plane is

$$x - 3y + 5z = 12$$

Solution:
If we let (x_0, y_0, z_0) be the point of intersection, then the coordinates of this point satisfy both the equation of the plane and the parametric equations of the line

$$x_0 - 3y_0 + 5z_0 = 12 \quad (a)$$

and for some value of t, say $t = t_0$

$$x_0 = 3 + 8t_0 \quad y_0 = 4 + 5t_0 \quad z_0 = -3 - t_0 \quad (b)$$

Substitute (b) in (a) yields

$$3 + 8t_0 - 3(4 + 5t_0) + 5(-3 - t_0) = 12$$

Solving for t_0

$$3 + 8t_0 - 12 - 15t_0 - 15 - 5t_0 - 12 = 0$$
$$8t_0 - 20t_0 - 12 - 12 - 15 + 3 = 0$$
$$-12t_0 = 36$$
$$t_0 = -3$$

Substitute $t_0 = -3$ into (b)

$$x_0 = 3 + 8(-3) \quad y_0 = 4 + 5(-3) \quad z = -3 - (-3)$$
$$(x_0, y_0, z_0) = (-21, -11, 0)$$

Example: Finding Intersection Between Line and Plane (2)

Find the intersection of the plane

$$2x - 8y + z = -77 \text{ and line } \vec{r}(t) = (-1, 2, -6)t + (-1, 6, -3)$$

Solution:
We have that

$$x(t) = -t - 1 \quad y(t) = 2t + 6 \quad z(t) = -6t - 3$$

We would like to find points x, y, z such that we have $-77 = -77$ with $x = 4$, $y = 3$, $z = -61$

$$2(4) - 8(3) - 61 = 8 - 24 - 61 = -77$$

Hence we have

$$\begin{aligned} \boldsymbol{p_0} &= (4, 3, -61) \\ \boldsymbol{n} &= (2, -8, 1) \\ \boldsymbol{v} &= (-1, 2, -6) \\ \boldsymbol{x_0} &= (-1, 6, -3) \end{aligned}$$

Recall that

$$\begin{aligned} (\boldsymbol{p} - \boldsymbol{p_0}) \cdot \boldsymbol{n} &= 0 \\ \text{Let } \boldsymbol{p} = \boldsymbol{r}(t) &= \boldsymbol{x_0} + t\boldsymbol{v} \end{aligned}$$

Hence we have

$$\begin{aligned} (\boldsymbol{x_0} + t\boldsymbol{v} - \boldsymbol{p_0}) \cdot \boldsymbol{n} &= 0 \\ [(\boldsymbol{x_0} - \boldsymbol{p_0}) + t\boldsymbol{v}] &= 0 \\ (\boldsymbol{x_0} - \boldsymbol{p_0}) \cdot \boldsymbol{n} + (t\boldsymbol{v}) \cdot \boldsymbol{n} &= 0 \\ (t\boldsymbol{v}) \cdot \boldsymbol{n} &= (\boldsymbol{p_0} - \boldsymbol{x_0}) \cdot \boldsymbol{n} \\ t &= \frac{(\boldsymbol{p_0} - \boldsymbol{x_0}) \cdot \boldsymbol{n}}{\boldsymbol{v} \cdot \boldsymbol{n}} = \frac{\boldsymbol{p} \cdot \boldsymbol{n}^T}{\boldsymbol{v} \cdot \boldsymbol{n}^T} \end{aligned}$$

So

$$\boldsymbol{p} = \boldsymbol{p_0} - \boldsymbol{x_0} = \begin{bmatrix} 4 - (-1) \\ 3 - (5) \\ -61 - (-3) \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ -58 \end{bmatrix}$$

$$\begin{aligned} \boldsymbol{p_0} \cdot \boldsymbol{n}^T &= (5, -3, -58) - (2, -8, 1)^T \\ &= 5(2) + (-3)(-8) + (-58) \\ &= 10 + 24 - 58 = 34 - 58 = -24 \end{aligned}$$

$$\begin{aligned} \boldsymbol{v} \cdot \boldsymbol{n}^T &= (-1, 2, -6) \cdot (2, -8, 1)^T \\ &= (-1)(2) + (2)(-8) + (-6)(1) \\ &= -2 - 16 - 6 = -24 \end{aligned}$$

$$
\begin{aligned}
t &= \frac{\boldsymbol{p} \cdot \boldsymbol{n}^T}{\boldsymbol{v} \cdot \boldsymbol{n}^T} = -24/-24 = 1 \\
\boldsymbol{p} &= \boldsymbol{r}(t) = \boldsymbol{x_0} + t\boldsymbol{v} \\
&= \begin{bmatrix} -1-1 \\ 6+2 \\ -3-6 \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \\ -9 \end{bmatrix}
\end{aligned}
$$

Example: Finding The t Parameter Given Normal, Points and Vector

Given these 4 coordinates, find the parameter t

- **normal**: $(2, -8, 1)$
- **point**: $(-2, 8, 9)$
- x_0: $(-1, 6, -3)$
- v: $(-1, 2, -6)$

Solution:

$$
\begin{aligned}
t &= \frac{(-\boldsymbol{x_0} + \boldsymbol{p_0}) \cdot \boldsymbol{n}}{\boldsymbol{v} \cdot \boldsymbol{n}} \\
&= \frac{(\boldsymbol{p_0} - \boldsymbol{x_0}) \cdot \boldsymbol{n}}{\boldsymbol{v_n} \cdot \boldsymbol{n}} \\
&= \frac{(-1, 2, -6) \cdot (2, -8, 1)}{(-1, 2, -6) \cdot (2, -8, 1)} \\
&= 1
\end{aligned}
$$

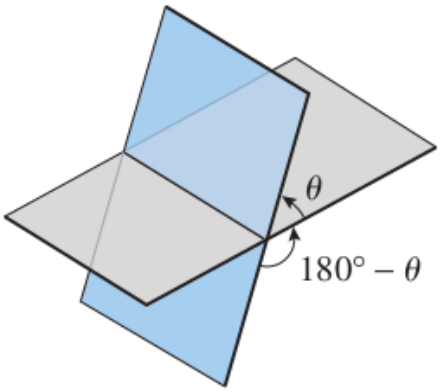
2.2.2 Intersecting Planes

Two distinct intersecting planes determine two positive angles of intersection-an (acute) angle θ that satisfies the condition $0 \leq \theta \leq \pi/2$ and the supplement of that angle. If $\mathbf{n}_1$ and $\mathbf{n}_2$ are normals to the planes, then depending on the directions of $\mathbf{n}_1$ and $\mathbf{n}_2$, the angle θ is either the angle between $\mathbf{n}_1$ and $\mathbf{n}_2$ or the angle between $\mathbf{n}_1$ and $-\mathbf{n}_2$. In both cases, the acute angle θ between the planes:

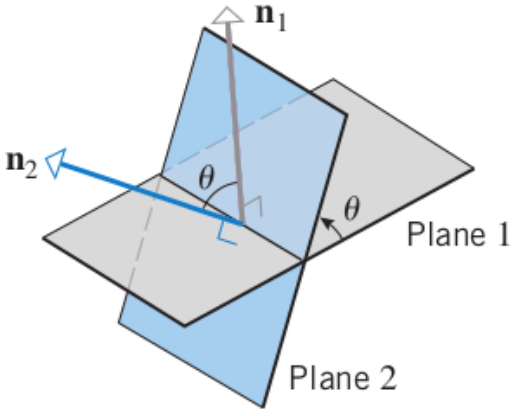
$$
\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \, \|\mathbf{n}_2\|}
$$

Example: Finding Angles of Intersection

Find the acute angle of intersection between the two planes.
Solution:



(a)



(b)

$$2x - 4y + 4z = 6 \text{ and } 6x + 2y - 3z = 4$$

Solution. The given equations yield the normals $\mathbf{n}_1 = \langle 2, -4, 4\rangle$ and $\mathbf{n}_2 = \langle 6, 2, -3\rangle$. Thus, Formula (9) yields

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\|\,\|\mathbf{n}_2\|} = \frac{|-8|}{\sqrt{36}\sqrt{49}} = \frac{4}{21}$$

from which we obtain

$$\theta = \cos^{-1}\left(\frac{4}{21}\right) \approx 79^\circ$$

Example: Finding Line Equation of Angles of Intersection

Find an equation for the line L of intersection of the planes in the previous example.
Solution.
First compute $\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2 = \langle 2, -4, 4\rangle \times \langle 6, 2, -3\rangle = \langle 4, 30, 28\rangle$. Since $\mathbf{v}$ is orthogonal to $\mathbf{n}_1$, it is parallel to the first plane, and since $\mathbf{v}$ is orthogonal to $\mathbf{n}_2$, it is parallel to the second plane. That is, $\mathbf{v}$ is parallel to L, the intersection of the two planes. To find a point on L we observe that L must intersect the xy-plane, $z = 0$, since

$\mathbf{v} \cdot \langle 0,0,1 \rangle = 28 \neq 0$. Substituting $z = 0$ in the equations of both planes yields

$$2x - 4y = 6$$
$$6x + 2y = 4$$

with solution $x = 1, y = -1$. Thus, $P(1,-1,0)$ is a point on L. A vector equation for L is

$$\langle x, y, z \rangle = \langle 1,-1,0 \rangle + t\langle 4,30,28 \rangle$$

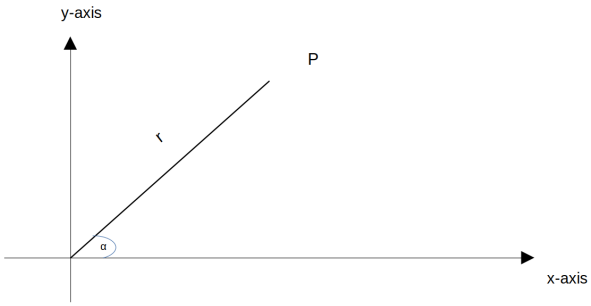
Chapter 3

Coordinatization

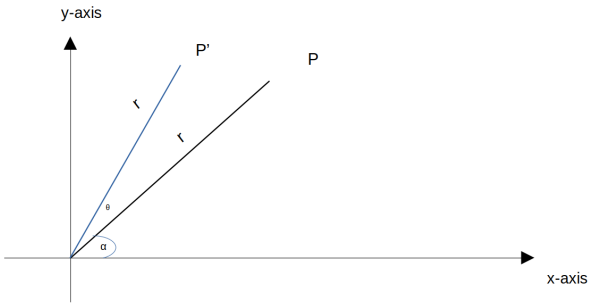
3.1 Coordinates

3.1.1 Change of Coordinates By Rotation

Let's look at the transformation that rotates a point by an angle. Let's take a simple example in $\mathbb{R}^2$ where we rotate the point P by an θ in the XY plane about the origin to get the point P'.



The coordinates of P are (x, y) and the coordinates of P' are (x', y').



We know that $x = r \cdot \cos\alpha$, and $y = r \cdot \sin\alpha$

$$
\begin{aligned}
x' &= r \cdot \cos(\alpha + \theta) \\
&= r(\cos\alpha\cos\theta - \sin\alpha\sin\theta) \\
&= r\cos\alpha \cdot \cos\theta - r\sin\alpha \cdot \sin\theta \\
&= x \cdot \cos\theta - y \cdot \sin\theta
\end{aligned}
$$

$$
\begin{aligned}
y' &= r\sin(\alpha + \theta) \\
&= r(\sin\alpha\cos\theta + \cos\alpha\sin\theta) \\
&= r\sin\alpha \cdot \cos\theta + r\cos\alpha \cdot \sin\theta \\
&= y \cdot \cos\theta + x \cdot \sin\theta
\end{aligned}
$$

$$
\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{\text{rotation matrix}} \cdot \begin{bmatrix} x \\ y \end{bmatrix}
$$

3.1.2 Extending To $\mathbb{R}^3$

We can easily extend the rotation transformation to $\mathbf{R}^3$. The transformation matrices for rotation in $\mathbf{R}^3$ about the standard X-axis, Y-axis, and Z-axis are as follows:

Rotation about Z-axis

$$
\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

Rotation about X-axis

$$
\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}
$$

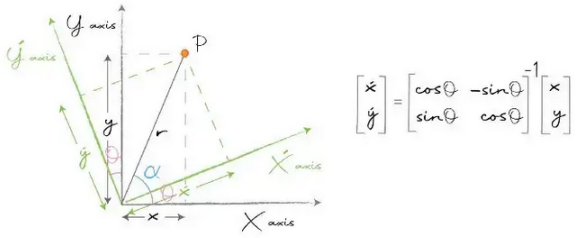
Rotation about Y-axis

$$
\begin{bmatrix} \cos\psi & 0 & -\sin\psi \\ 0 & 1 & 0 \\ \sin\psi & 0 & \cos\psi \end{bmatrix}
$$

3.1.3 Change of Basis Transformation

A change of basis matrix is the **inverse** of the linear transformation matrix. This means that if we know the camera transformation matrix, then we can take its inverse to get the change of basis matrix which can help us find the coordinates of points with respect to the camera.

The goal is to find the coordinates of a point with respect to a new basis. The point will be fixed.

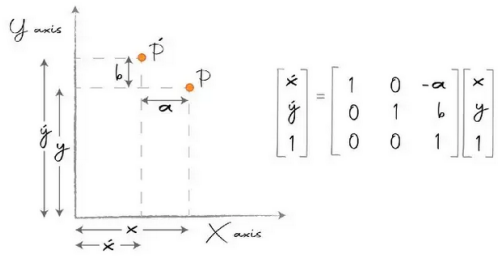


Rotating the XY axes

$$\begin{bmatrix} x\prime \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & +\cos\theta \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$$

3.1.4 Change of Coordinates By Translation

The idea of translation is simple — given a point P, we move it by an offset to get the point P' as shown in the below figure:



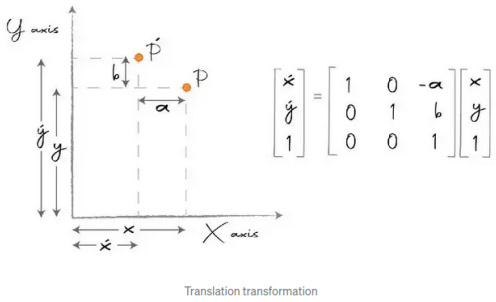
Translation transformation

The coordinates represented with an extra dimension are called homogeneous coordinates. To get back the euclidean coordinates from the homogeneous coordinates, we simply divide by the last element like so:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{x}{1} \\ \\ \frac{y}{1} \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

3.1.5 Change of Basis By Translation

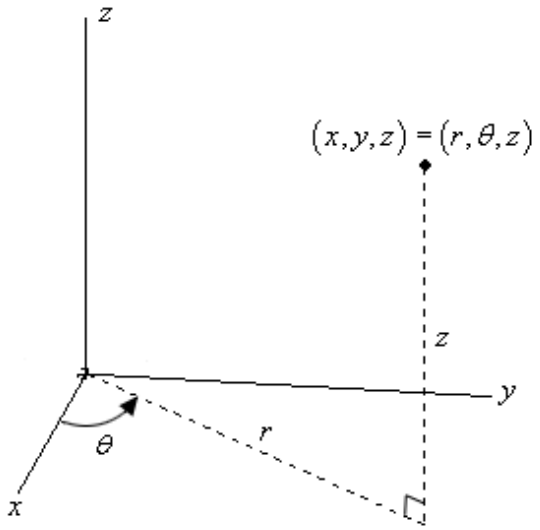
Like we saw earlier, In a change of basis transformation we transform the axes rather than the points. In the below example, we move the XY axes by an offset to get X' Y'. Our goal is to find the coordinates of point P wrt the new axes X' Y'.



The coordinates of P wrt the old axes XY are (x, y) and wrt to the new axes, X'Y' are (x',y'). The offset here is (a, b)

$$\begin{bmatrix} x\prime \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -a \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3.1.6 Cylindrical Coordinates



Cylindrical coordinates are an extension of polar coordinates in $\mathbb{R}^3$ such that a point in cartesian coordinates can be represented as follows.

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

A point in cylindrical coordinates can be found:

$$r = \sqrt{x^2 + y^2} or r^2 = x^2 + y^2$$
$$\theta = \tan^{-1}(\frac{y}{x})$$
$$z = z$$

Example: Convert Cartesian Coordinates To Cylindrical Coordinates

Convert the rectangular coordinates $(1, -3, 5)$ to cylindrical coordinates. **Solution:** Use the second set of equations from Note to translate from rectangular to cylindrical coordinates:

$$\begin{aligned} r^2 &= x^2 + y^2 \\ r &= \pm\sqrt{1^2 + (-3)^2} \\ &= \pm\sqrt{10}. \end{aligned}$$

We choose the positive square root, so $r = \sqrt{10}$. Now, we apply the formula to find θ. In this case, y is negative and x is positive, which means we must select the value of θ between $\frac{3\pi}{2}$ and 2π

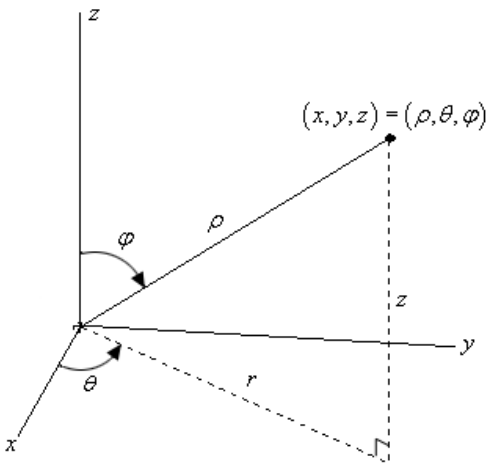
$$\begin{aligned} \tan\theta &= \tfrac{y}{x} & &= \tfrac{-3}{1} \\ \theta &= \arctan(-3) & &\approx 5.03\text{rad} \end{aligned}$$

In this case, the z-coordinates are the same in both rectangular and cylindrical coordinates:

$$z = 5$$

The point with rectangular coordinates $(1, -3, 5)$ has cylindrical coordinates approximately equal to $(\sqrt{10}, 5.03, 5)$.

3.1.7 Spherical Coordinates

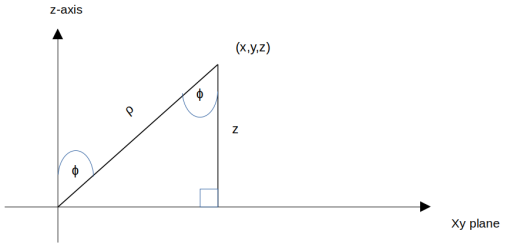


ρ is the distance from the origin to the point and will require $\rho \geq 0$
θ is the angle between the positive $x-axis$ and the line above denoted by r (some r in polar/cylindrical coordinates)
ϕ is the angle between positive z-axis and the line from the origin to the point. We will require $0 \leq \phi \leq \pi$

In summary, ρ is the distance from the origin to the point, ϕ is the angle that we need to rotate down from the positive z-axis to get to the point and θ is how much we need to rotate around the z-axis to get to the point.

Let's start with a point in spherical coordinates and ask what the original coordinates of the point are.

So, we know ρ, θ, ϕ and we want to find (r, θ, z).



$$z = \rho \cdot cos\phi$$
$$r = \rho \cdot sin\phi$$
$$\theta = \theta$$

Pythagorean Theorem

$$\rho^2 = r^2 + z^2$$

Next, we find the cartesian coordinates of the same point

$$x = rcos\theta$$
$$y = rsin\theta$$
$$z = z$$

$$\therefore x = \rho sin\phi cos\theta$$
$$y = \rho sin\phi sin\theta$$
$$z = \rho cos\phi$$

$$\text{Since} r^2 = x^2 + y^2, \text{ then}$$
$$\rho^2 = x^2 + y^2 + z^2$$

Example: Convert From Cylindrical Coordnates To Spherical Coordinates

Convert $(\sqrt{6}, \frac{\pi}{4}, \sqrt{2})$ from cylindrical coordinates to spherical coordinates.
Solution:
θ is the same in both coordinate systems. Let's find ρ

$$\rho = \sqrt{r^2 + z^2} \qquad = \sqrt{7} = 2\sqrt{2}$$

Fially, let's get ϕ

$$z = p \cdot cos\phi$$
$$cos\phi = \frac{z}{\rho} \qquad = \frac{\sqrt{2}}{2\sqrt{2}}$$
$$\phi = cos^{-1}(\frac{1}{2}) = \frac{\pi}{3}$$

Hence, the spherical coordinates are

$$(\rho, \theta, \phi) = (2\sqrt{2}, \frac{\pi}{4}, \frac{\pi}{3})$$

Example: Convert From Cartesian Coordinates to Spherical Coordinates

Convert $(-1, 1, -\sqrt{2})$ from cylindrical coordinates to spherical coordinates.
Solution:
Find ρ,

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + 1 + 2} = 2$$

Now, we find ϕ

$$z = \rho cos\phi$$
$$cos\phi = \frac{z}{\rho} = \frac{-\sqrt{2}}{2}$$
$$\phi = \frac{3\pi}{4}$$

Find theta

$$y = r\sin\theta \text{ where } r = \rho\sin\theta = \frac{2\sqrt{2}}{2}$$

$$\sin\theta = \frac{y}{r} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\implies \theta = \frac{\pi}{4}$$

$$\because sin \to \theta = n \cdot \pi + (-1)^n \cdot a$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{3 \cdot \pi}{4}$$

Point(s) with coordinates $x = -1, y = 1$ lies in the second quadrant hence the second θ is $\frac{3 \cdot \pi}{4}$. Thus the spherical coordinates are $(2, \frac{3\pi}{4}, \frac{3\pi}{4})$

Chapter 4

Cross Product

Definition 4.0.1 (Definition 1)**.** If $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ are vectors in 3-space, then the cross product $\mathbf{u} \times \mathbf{v}$ is the vector defined by

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$$

or, in determinant notation,

$$\mathbf{u} \times \mathbf{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right)$$

Example: Calculating A Cross Product
Find $\boldsymbol{u} \times \boldsymbol{v}$ where $\boldsymbol{u} = (1, 2, -2)$ and $\boldsymbol{v} = (3, 0, 1)$ **Solution**

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \left(\begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix}, - \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} \right) \\ &= (2, -7, -6) \end{aligned}$$

Theorem 5 (Cross Product)**.** *Relationships Involving Cross Product and Dot Product If* $\mathbf{u}, \mathbf{v}$*, and* $\mathbf{w}$ *are vectors in 3-space, then*
(a) $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$ *[*$u \times$v *is orthogonal to* u*]*
(b) $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0$ [u $\times$ v *is orthogonal to* v]
(c) $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$ *[Lagrange's identity]*
(d) $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ *[vector triple product]*
(e) $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{v} \cdot \mathbf{w})\mathbf{u}$ *[vector triple product*]

Example: $u \times v$ **is perpendicular to** u **and to** v
Consider the vectors

$$\mathbf{u} = (1, 2, -2) \quad \text{and} \quad \mathbf{v} = (3, 0, 1)$$

In Example 1 we showed that

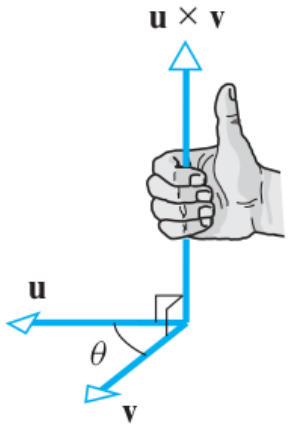
$$\mathbf{u} \times \mathbf{v} = (2, -7, -6)$$

Since

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = (1)(2) + (2)(-7) + (-2)(-6) = 0$$

and

$$\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = (3)(2) + (0)(-7) + (1)(-6) = 0$$



Theorem 6 (Properties of Cross Product)**.** *Properties of Cross Product If* $\mathbf{u}, \mathbf{v}$*, and* $\mathbf{w}$ *are any vectors in 3-space and* k *is any scalar, then:*
(a) $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
(b) $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
(c) $(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$
(d) $k(\mathbf{u} \times \mathbf{v}) = (k\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (k\mathbf{v})$
(e) $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$
(f) $\mathbf{u} \times \mathbf{u} = \mathbf{0}$

Definition 4.0.2 (Determinant Form of Cross Product)**.**

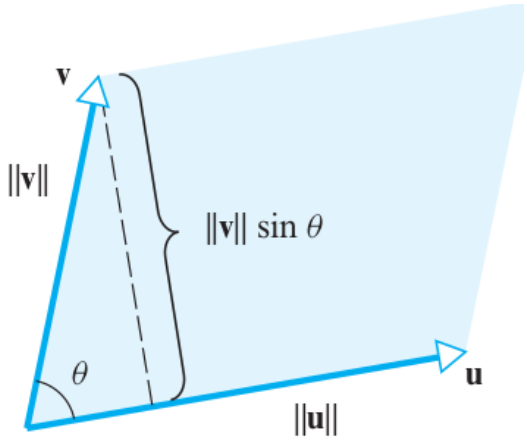
$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}$$

For example, if $\mathbf{u} = (1, 2, -2)$ and $\mathbf{v} = (3, 0, 1)$, then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -2 \\ 3 & 0 & 1 \end{vmatrix} = 2\mathbf{i} - 7\mathbf{j} - 6\mathbf{k}$$

Geometric Interpretation of Cross Product

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in 3-space, then the norm of $\mathbf{u} \times \mathbf{v}$ has a useful geometric



interpretation. Lagrange's identity, given in Theorem 3.5.1, states that

$$\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$$

If θ denotes the angle between $\mathbf{u}$ and $\mathbf{v}$, then $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$, so (5) can be rewritten as

$$\begin{aligned}\|\mathbf{u} \times \mathbf{v}\|^2 &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \cos^2\theta \\ &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \left(1 - \cos^2\theta\right) \\ &= \|\mathbf{u}\|^2\|\mathbf{v}\|^2 \sin^2\theta\end{aligned}$$

Since $0 \leq \theta \leq \pi$, it follows that $\sin\theta \geq 0$, so this can be rewritten as

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin\theta$$